Intelligent Security System for Women & Child Safety

A Project Report submitted in partial fulfillment of the requirements for award of the degree

BACHELOR OF ENGINEERING

in

ELECTRONICS & TELECOMMUNICATION ENGINEERING

of

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

by

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Certificate

This is to certify that the Project Report entitled

Intelligent Security System for Women & Child Safety

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We are feeling very humble in expressing my gratitude. It will be unfair to bind the precious help and support which we got from many people in few words. But words are the only media of expressing one's feelings and my feelings gratitude is absolutely beyond these words. It would be my pride to take this opportunity to say the thanks.

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Abstract

Now-a-days we see that the cases of women harassment are increasing day by day. In fact we've heard about the rape cases of small children also. Also the kidnapping of small children are also increasing day by day in the rural area. The idea behind our project is to give the security to the women as well as child whom so ever will get harassed, it will detect the heartbeat, temperature, etc.& if the limit will get exceed above a certain level then the location of the subject will get tracked & it will send a message with the location of the subject to the contacts fed in the programs such as relatives or control room, etc. using GSM. With this we had an additional feature of security of blind person also. If a blind person is walking in the crowded place or on road & if any obstacle gets detected within a certain range in front of the Blind person, then the alarm will get activated & will inform the person about the obstacle.

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1. Introduction

1.1.Introduction :

Women safety has become a matter of utmost concern in the current times. Most cases of women harassment occur due to the lack of proper help at the right time to the victim. With the advancement of technology many systems has evolved that serve as an aide to women in dangerous situations. As the cases of women harassment are increasing day by day, so we tried to build a device which can be useful for a woman in distress. Although there exists many security systems, an advanced smart systems is a necessity. This project proposes a smart security device to help women to be safe during times of utmost danger.

The device comprises of an Arduino UNO board, various sensors to monitor the body parameters variations, GPS & GSM modules. When a device notices an abnormal variations in the body parameters the distress message with location coordinates is sent to registered contacts.

This device also works for the Blind Person Safety. For this we've used an ultrasonic sensor which will alert the person if any obstacle is detected in front of the person by triggering an alarm.

1.2.Relevance / Significance :

Various apps & devices are designed for the safety of Women & Child & also for the blind person. As per the survey in Sept. 2020, India recorded an average of 87 rape cases daily and overall 4,05,861 cases of crime against women during the year, which is a rise of over 7% from 2018. So, safety has become a very important aspect for the women & child now-a-days.

1.3.Specification :

1.3.1. Electrical Specification :

1) 9V battery
2) GPS Location Tracking
3) GSM based message sending

1.3.2. Physical Specification :

1) Weight : 0.5 kg (approx.)
2) Height : 7 cm
3) Length : 17 cm
4) Range of Operation : 4.5 m

1.4. Platform Used :

1.4.1. Hardware:

1) PCB
2) Plastic Box

1.4.2. Software :

1) Arduino IDE
2) Proteus

1.5.Advantages :

1) The main advantage of this system is that this device is small & easy to carry.
2) The sensor has excellent sensitivity combined with a quick fast response time.
3) The system is highly reliable, tamper-proof and secure.
4) It is possible to get instantaneous results and with high accuracy.

*1.6.*Applications :

1) This project is used for Women's Safety as women harassment cases are increasing rapidly. So the woman can carry this device with her while going outside
2) Also, this project aims at child's safety as kidnapping of many small kids is increasing day by day.
3) Can be used for the safety of elderly aged people.
4) Also this project is for the blind person which will inform them about the harm.

2. Literature Review

2.1. Recent Papers :

An extensive overview of the literature is mentioned in this section.

2.1.1. Intelligent Security Systems for Women by using Arduino :

"Intelligent Security Systems for Women by using Arduino" by Mr. V. Nanthakumar, Ms. G. Hariprabha, Ms. D. Jasmine, Ms. B. Keerthana, Department of EEE, Sengunthar Engineering College, Tiruchengode, Tamil Nadu, India dated 2019. This paper introduces a new device, security system specially designed for women in distress. Women can press the button that is attached to the device and the location information is sent as an SMS alert to few pre-defined emergency numbers in terms of latitude and longitude.

2.1.2. Design and Implementation of Women Safety System Based On IoT Technology :

"Design and Implementation of Women Safety System Based On IoT Technology" by B. Sathyasri, U. Jaishree Vidhya, G. V. K. Jothi Sree, T. Pratheeba, K. Ragapriya, Department of Electronics and Communication Engineering, Vel Tech, Chennai, India dated 2019. This paper covers descriptive details about the design and implementation of "Smart Band". In this project when a woman senses danger she has to hold ON the trigger of the device. Once the device is activated, it tracks the current location using GPS and sends emergency message using GSM to the registered mobile number and near by the police station. IoT module is used to track the location continuously and update into the webpage. The main advantage of this project is that this device can be carried everywhere since it is very small.

2.1.3. Women Safety System using Arduino UNO & Integrated Safety App :

"Women Safety System using Arduino UNO and Integrated Safety App" by R Anitha, Valliammai Engineering College, Kanchipuram, Tamil Nadu, India. This paper discussed a device called as "Virtual Friend", specially designed for women in trouble. The basic approach is to use the Arduino UNO microcontroller based on ATmega328P has the function of send and receive data which is provided by Arduino GSM shield using GSM network. Arduino UNO gets the coordinates of the current location; it transfers coordinate details to the user's mobile via Arduino GSM module. The SOS light is a signal used to alert the passerby and it gives the sign of universal help to the victim who is in distress. The alarm buzzer is activated if the woman is in danger situation. In the critical situation, the woman send the message or make a call including the location of the particular incident to the registered mobile numbers through the use of GSM and GPS.

2.2. Literature Survey :

The above all papers are given in the table below:

Table No. 2.1 Literature Survey

Sr.No.	Name of Project	Advantages
1.	Intelligent Security Systems for Women by using Arduino	It informs about the situation using text message with her GPS location due to which instantly she gets the help.
2.	Design and Implementation of Women Safety System Based On IoT Technology	This system starts the alarm due to which the people in the surrounding should be informed & get instant help.
3.	Women Safety System using Arduino UNO and Integrated Safety App	As there is app with this system, it is very small in size & can be easy carried.

2.3. Study of Components :

2.3.1. Controller [Arduino UNO (ATmega 328P)] :

The microcontroller board like "Arduino Mega" depends on the ATmega2560 microcontroller. It includes digital input/output pins-54, where 16 pins are analog inputs, 14 are used like PWM outputs hardware serial ports (UARTs) – 4, a crystal oscillator-16 MHz, an ICSP header, a power jack, a USB connection, as well as an RST button. This board mainly includes everything which is essential for supporting the microcontroller.

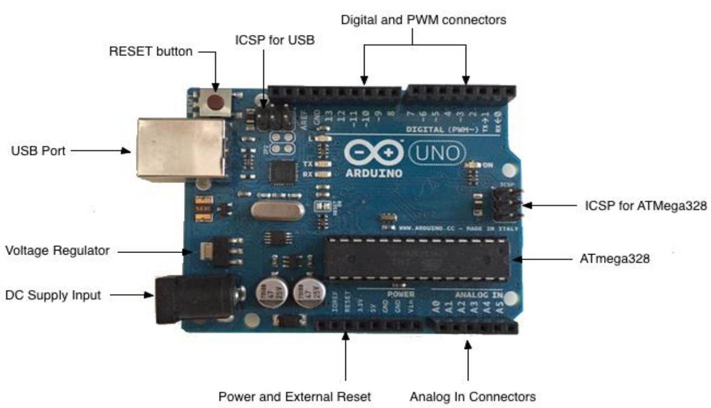

Fig 2.1 Arduino UNO Board

2.3.2. Body Temperature Sensor (DS18B20) :

DS18B20 is a temperature sensor which has stainless steel tube encapsulation which makes it waterproof, whose output voltage varies, based on the temperature around it. It is a small rod like structure with cable which can be used to measure temperature anywhere between -55°C to 125°C. It can be easily interfaced with any Microcontroller that has ADC function or any development platform like Arduino. As it has Stainless probe, it gives accurate measurement & have high efficiency.

Fig 2.2 DS18B20 Body Temperature Sensor

2.3.3. Heartbeat sensor (Finger Based) :

Heartbeat Sensor is an electronic device that is used to measure the heart rate i.e. speed of the heartbeat. Monitoring body temperature, heart rate and blood pressure are the basic things that we do in order to keep us healthy. Heart Rate can be monitored in two ways: one way is to manually check the pulse either at wrists or neck and the other way is to use a Heartbeat Sensor. The principle behind the working of the Heartbeat Sensor is Photoplethysmograph. According to this principle, the change in the volume of blood in an organ is measured by the changes in the intensity of the light passing through that organ.

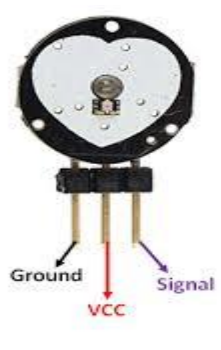

Fig 2.3 Heartbeat Sensor

2.3.4. **Ultrasonic Sensor (HC-SR04) :**

An Ultrasonic sensor is an electronic device that measures the distance of a target object by emitting ultrasonic sound waves, and converts the reflected sound into an electrical signal. Ultrasonic waves travel faster than the speed of audible sound (i.e the sound that humans can hear). The HC-SR04 ultrasonic sensor uses SONAR to determine the distance of an object just like the bats do. It offers excellent non contact range detection with high accuracy and stable readings in an easy-to-use package from 2 cm to 400 cm.

Fig 2.4 HC-SR04 Ultrasonic Sensor Module

2.3.5. **Buzzer :**

A buzzer is an audio signaling device & it is of piezoelectric type. Piezoelectric buzzers are sounding bodies that use unimorph piezoelectric vibration plates and feature small sizes, low voltages, and loud sound volumes, as well as high performance and a design that allows for easy mounting on circuits. Since these buzzers are separately-excited oscillation, they accommodate flexible designs as music tone speakers or buzzers.

Fig 2.5 Piezoelectric Buzzer

3. Design and Development

3.1. Block Diagram and Description :

The design of the hardware system is crucial in bringing the mechanism and software to work together. The main components in the circuit of Intelligent Security System for Women & Child Safety are Temperature Sensor, Ultrasonic Sensor, Heartbeat Sensor, GSM Module, GPS Module and Arduino UNO (ATMega328P) Controller. Figure 9. Shows the overall block diagram of the electronic system for the Intelligent Security System for Women & Child Safety. All the sensors are used the detect the various parameters & if any 2 sensors value crosses the threshold then the text message will be sent to the contact number feed in the project with the GPS Location of the subject.

Figure 10. Shows the block diagram of the electronic system for the Blind Peron Safety. In this Ultrasonic sensor is used to detect the obstacle in front of the. If the threat is detected then the blind person will be informed by activating the Alarm.

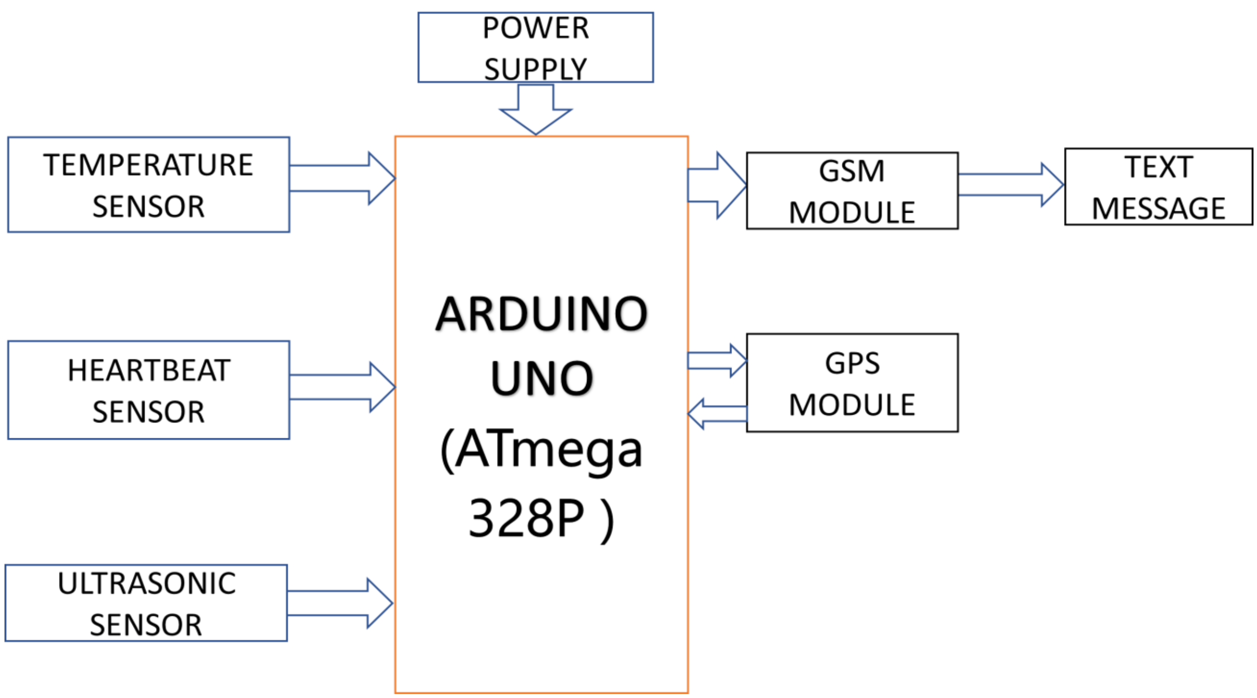

Fig 3.1 Block Diagram for Women & Child Safety

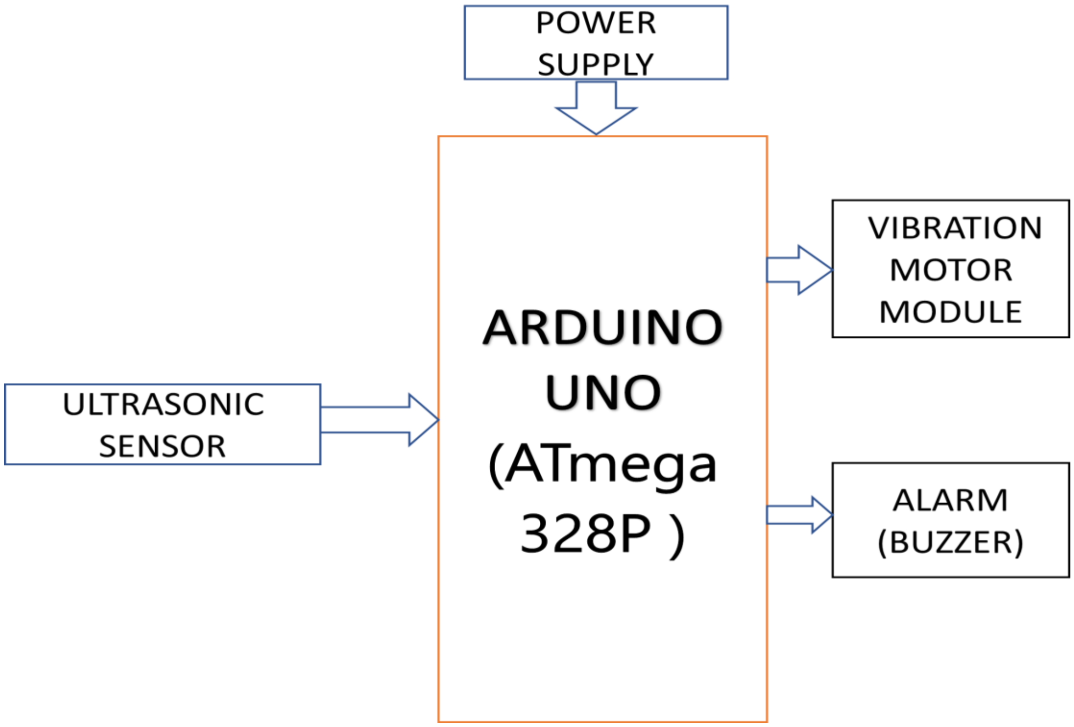

Fig 3.2 Block Diagram for Blind Person Safety

3.2. Circuit Diagram :

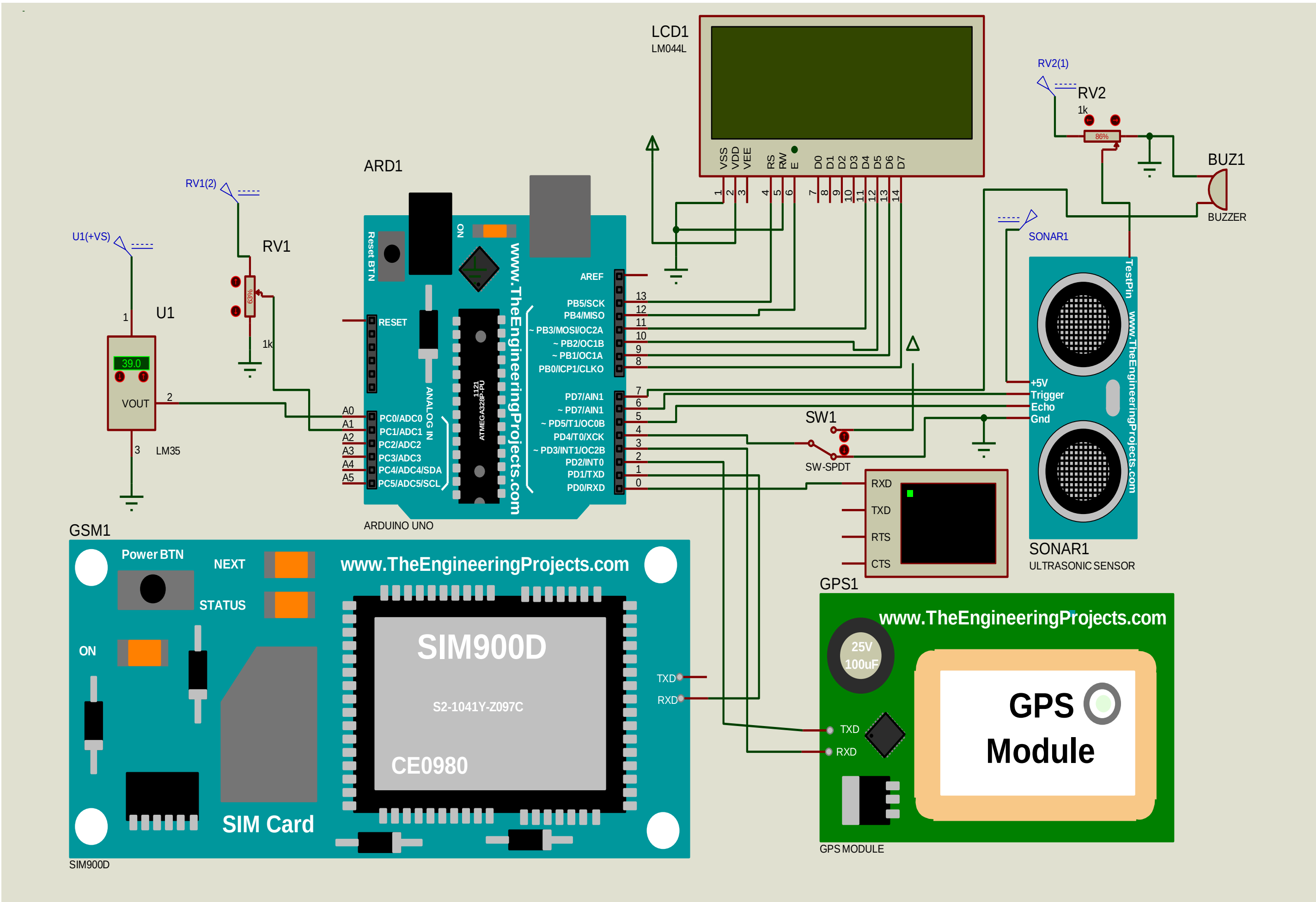

Fig 3.3 Circuit Diagram for Intelligent Security System for Women & Child Safety

3.3. Software Design Steps :

3.3.1. For Women & Child Safety :

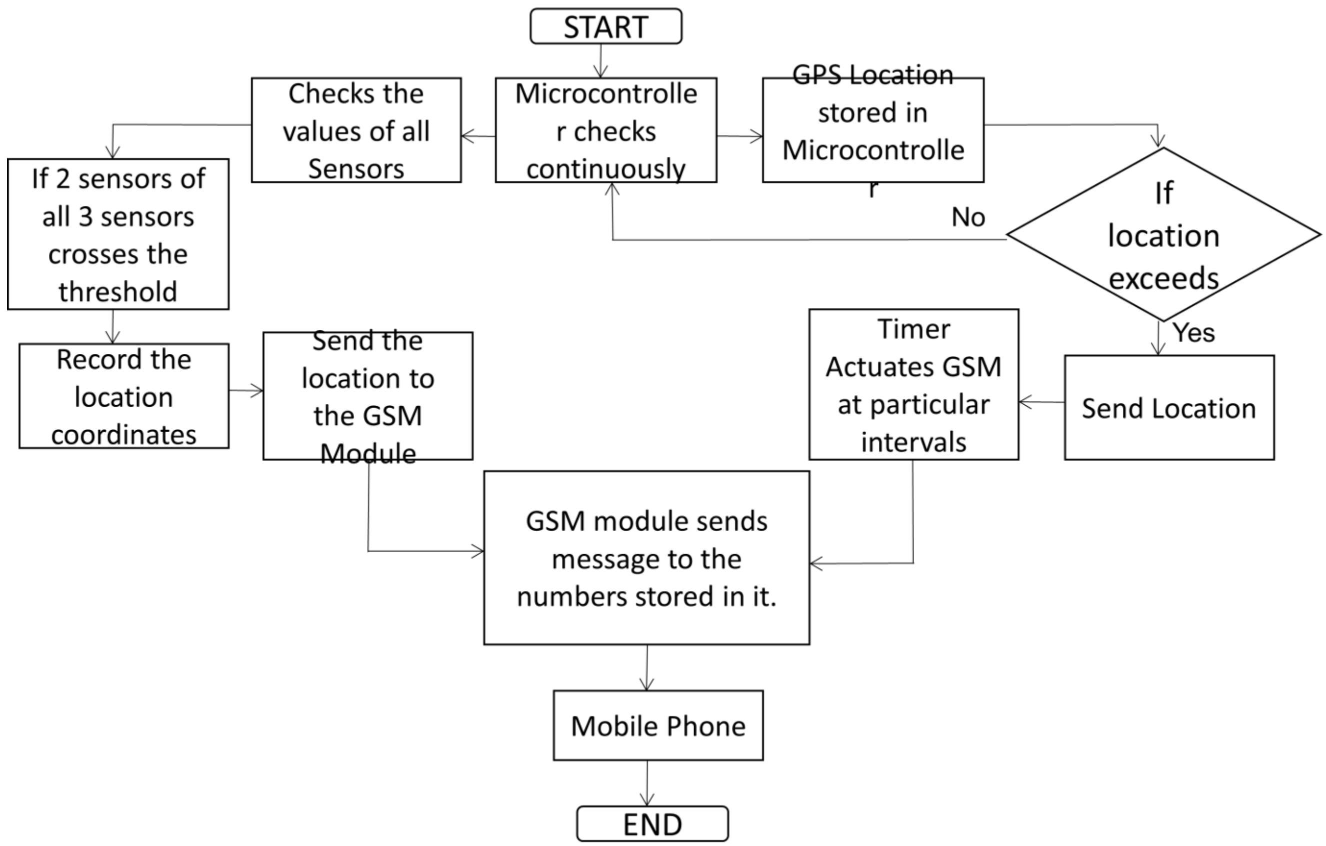

Fig 3.4 Software Design steps for Women & Child Safety

3.3.2. For Blind Person Safety :

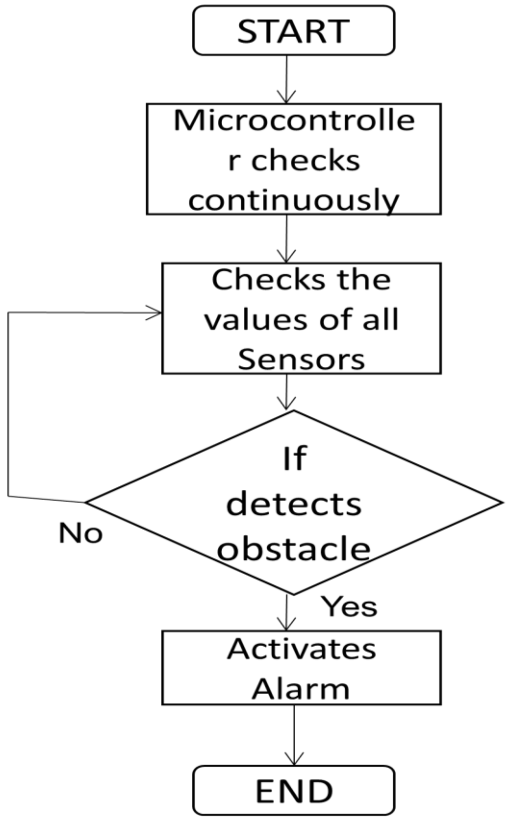

Fig 3.5 Software Design steps for Blind Person Safety

When the power supply of the project is switch & if the switch of the project is in ON state (Women & Child Safety Mode), then all 3 sensors starts checking the value and the GPS & GSM Modem gets initialized. Controller will continuously receive the commands from all the sensors. Let's assume that a women / child is getting harassed, then the parameters such as temperature, heartbeat, etc. gets increases. If any of the 2 sensors detects the value above the threshold then a signal is sent form the controller to the GSM & a text message is sent to the contact number which is feed in the project i.e relative, control room, etc.

If the switch of the project is changed from ON state to OFF state then the project will start working in Blind Person Safety Mode. Then the Ultrasonic sensor will continuously senses the parameter & if the value crosses the threshold, then the controller will send the signal to the Alarm. If an obstacle is detected, Alarm will be activated & the person will be informed about the harm.

3.4. PCB DESIGNING AND FABRICATION

DESCRIPTION OF PRINTED CIRCUIT BOARD

PCB designing and fabrication

It is called PCB in short printed circuit pattern applied to one or both sides of an insulating base, depending upon that, and it is called single sided PCB or double-sided PCB. Conductor materials available are silver, brass, aluminum and copper; copper is most widely used which is used here as well. The thickness ofconducting material depends on the current carrying capacity of the circuit.

The printed circuit board usually serves three functions:

1. It provides mechanical support to the components mounted on it.
2. It provides necessary electrical interconnection.
3. It acts as heat sink i.e., it provides a conduction path leading to removal of most of the heat generated in thecircuit.

Cu clad

The base of laminate is either paper of glass fiber cloth. Cu foil, which is produced by the method of electroplating, is placed on laminate and both are kept under hydraulic pressure for proper adhesive pressure forproper adhesive. These Cu clad are easily available in the market.

Types of Laminates

National Electrical Manufactures Association (NEMA) has various grades of laminates that are obtained by different resins and filters.

1. Phenol

Phenol and Formaldehyde produce phenolic paper base laminate it has phenolic resins with proper filter. This is Brown in color and opaque. Disadvantage is poor moisture resistance.

2. Epoxy Laminates

Epoxy paper that is also paper based but impregnated with epoxy resin, yellowish white and translucent.

Epoxy Glass; this base material has high mechanical strength and good electrical properties usually green in color and semitransparent.

There are a variety of laminates available. We have selected Fiber Glass epoxy laminate. PCB fabrication includes following steps:

i. Layout of the circuit

ii. Artwork designing

iii. Printing

iv. Etching

v. Drilling

vi. Mounting of components and soldering

vii. Finishing

- **Layout**

The layout of a PCB has to incorporate all the board before one can go onto the all work preparation. Detailedcircuit diagram, the design concept and the philosophy behind the equipment are very important for the layout.

Layout Scale-

Depending on the accuracy required artwork should be produced at a 1:1 or 2:1 or even 4:1 scale. The layout is best prepared on the same scale as the artwork to prevent the entire problem, which might be caused by redrawing of the layout to the artwork scale. The layout/ artwork scale commonly applied is 2:1 with a 1:1 scale, no demanding single sided boards can be designed but sufficient care should be taken, particularly during the artwork preparation.

Procedure

The first rule is to replace each and every PCB layout as viewed from the component side. This rule must be strictly followed to avoid confusion, which would otherwise be caused.

Another important rule is not to start the designing of a layout unless an absolutely clear circuit diagram is available.

Among the components, the larger ones are placed first and the space in between is filled with smaller ones. Components requiring input/output connecting come near the connector. All components are placed in such a manner that de-soldering of other components is not necessary if they have to be replaced.

Layout sketch

The end product of the layout designing is the pencil sketched component and conductor drawing which is caller 'layout sketched'. It contains all the information for the preparation of the network.

Component holes

In a given, PCB most all the holes required are one particular diameter. Holes of a different are shown with a code in the actual layout sketch.

Conductor Holes

A code can be used for the conductor with a special width. Minimum spacing should also be provided.

A) Holes	B) Conductor Widths
Standard holes	Standard width, 0.5 mm
1.1 mm	1 mm
1.5 mm	2 mm
3.2 mm	4 mm

Artwork

The generation of PCB artwork should be considered as the first step of the PCB manufacturing process. The importance of a prefect artwork should not be under estimated. Problems like inaccurate registered, broken annular rings or too critical spacing are often due to bad artwork. And even with the most sophisticated PCB production facilities, PCB can be made better than the quality of the artwork used.

Basic Approaches

For ink drawing on white cardboard paper, good quality Indian ink and ink-pen set are minimum requirements. Drawing practice-drawing procedure is very at-least by 0.1 – 0.2, and solder pad locations. And conductors can be easily displaced by 0.3 – 0.5 mm

Screen Printing

The process of screening – printing is well known to the printing industry because of its inherent capabilities of printing a wide range of inks on almost any kind of surface including glass, metal, plastic fabrics etc.

Found their way into an extremely broad field of applications.

Screen-printing offers the advantage of wide control on the ink deposition, thickness though the selection of suitable mass density and composition. In the production of PCB’s, it is successfully employed in printing of

- Etch resists
- Plate resists
- Solder stop lacquers
- Notation printing

In its basic form, the screen-printing process is very simple. A screen fabric with uniform meshes and opening is stretched and fixed on a solid frame of metal or wood. The circuit pattern area gets open, while the meshes in the rest of the area are closed.

In the actual printing step, ink is forced by the moving squeeze through the open meshes onto the surface of the material to be printed.

The ink deposition, in a magnified cross section, shows the shape of a trapezoid.

Pattern transfer onto the Screen

There are two different methods in use, and each method has its own advantages and disadvantages.

With the direct method, the screen is prepared by coating a photographic emulsion directly onto the screen fabric and exposing it in the pattern area. The indirect method makes use of a separate screen process film, supported on a backing sheet. The film on its backing sheet that is there after pressed onto the screen fabric and sticks there. Finally, the backing sheet is peeled off, opening all those screen meshes, which are not covered by the film pattern.

The direct method provides very durable screen stencils with a higher dimensional accuracy but the finest details are not reproduced. The indirect method is more suitable for smaller series and where the finest details to be reproduced. The indirect method is faster but dimensionally less accurate and the screen stencils are less durable, more sensitive to mechanical damages and interruption in printing.

Etching

In all subtractive PCB process, etching is one of the most important steps. The final copper pattern is formed byselective removal of all the unwanted copper, which is not protected by an etching unit.

Solutions, which are used in etching process, are known as enchants.

I) Ferric Chloride

II) Cupric Chloride

III) Chromic Acid

IV) Alkaline Ammonia.

Of these Ferric Chloride is widely used because it has short etching time and it can be stored for a long time. Etching of PCBs as required in modern electronic equipment production, is usually done in spray type etching machines.

Tank or bubble etching, in which the boards kept in tank, were lowered and fully immersed into the agitated,has almost disappeared.

Component Mounting

Careful mounting of components on PCB increases the reliability of assembly.

1) The leads must be cleaned before they are inserted in PCB holes. Asymmetric lead bending must beavoided; the ENT leads must fit into holes properly so that they can be soldered.
2) When space is to be saved then vertical mounting is to be preferred. The vertical leads must have aninsulating sleeve.
3) Where jumper wire crosses over conductors, they must be insulated.
4) For mounting of PCBs, TO5, DIP packages special jigs must be used of easy insertion.
5) While mounting transistors, each lead must insulating sleeve. All the flat radial components such as resistors, diodes and inductors are mounted and soldered. Then IC bases are soldered. The vertical components such as transistors, gang condenser and FET are mounted & soldered.

Soldering

The next process after the component mounting is soldering; solder pint is achieved by heating the solder and base metal about the melting point of the solders used.

The necessary heat depends upon:

1) The nature and type of joints
2) Melting temperature of solder
3) Flux

Soldering techniques are of so many types but we are using iron soldering.

Iron soldering

Soldering iron consists of an insulating handle connected through a metal shaft, os a bit accurately makes contact with the component parts of the joint and solder and heats them up. The electrical heating element is located in the hollow shank with the handle to heat the bit.

Functions of the Bit

It stores the heat and conveys it from the heat source to the work. It may be required to store surplus solder fromthe joint. It may be required to store molten solder and flux to the work.

The surface must be lined and wetted; this encourages flow of solder into the joint. When the surface of the work becomes tested by the solder, a continuous flow of liquid metal between the bit and the work provides a path of high thermal conductivity through which heat can flow into the work piece.

Solder bit are made up of copper; this metal has good wetting properly, heat capability and thermal conductivity. Tin-lead solder affects copper during soldering operation. Production of copper bit can be made with thick iron coating followed by Ni/Tin plating. The life of the bit is increased by a factor of 10 to 15. Solderirons are specified in terms of wattage. Depending on heat input intended for working and types of work (continuous or individual) the choice of the solder iron can be made.

Procedure of Soldering

The points to be joined must be cleaned first and fluxed. The hard solder iron and solder wire is applied to the work. The melted solder becomes bright and fluid. The iron must be removed after sufficient time and joint is allowed to coal.

At the end, finishing is done.

PCB designing using computer aided designing (CAD):

CAD has many advantages over manual designing, important among them is:

1) Changes can be easily made because we don’t have to erase our pencil work on paper repeatedly.
2) Time is saved.
3) Before taking printout we can have preview of the design etc.

The software which we have used is Quick-route.

3.5. Bill of Material

Table No. 3.1 Bill of Materials

Sr. No.	Components	Qty.	Cost
1.	Arduino UNO Board	1	559/-
2.	GPS Module	1	650/-
3.	GSM Module	1	1190/-
4.	Ultrasonic Sensor	1	105/-
5.	Body Temperature Sensor (DS18B20)	1	273/-
6.	Heartbeat Sensor (Finger based)	1	263/-
7.	Potentiometer (10kΩ)	2	56/-
8.	Buzzer	1	20/-
9.	SPDT Switch	1	10/-
10.	Transformer primary 230volt AC,secondary,0-12.1AMP	1	150/-
11.	P-N Junction diode (1N4007)	4	4/-
12.	Cp-1000MFD/25volt electrolytic capacitor	1	8/-
13.	IC4-7805 Positive voltage regulator(+5volt)	1	15/-
14.	Resistor (680Ω)	1	1/-

4. Result and Discussion

4.1. Simulation Result :

4.1.1. When Project is turned ON :

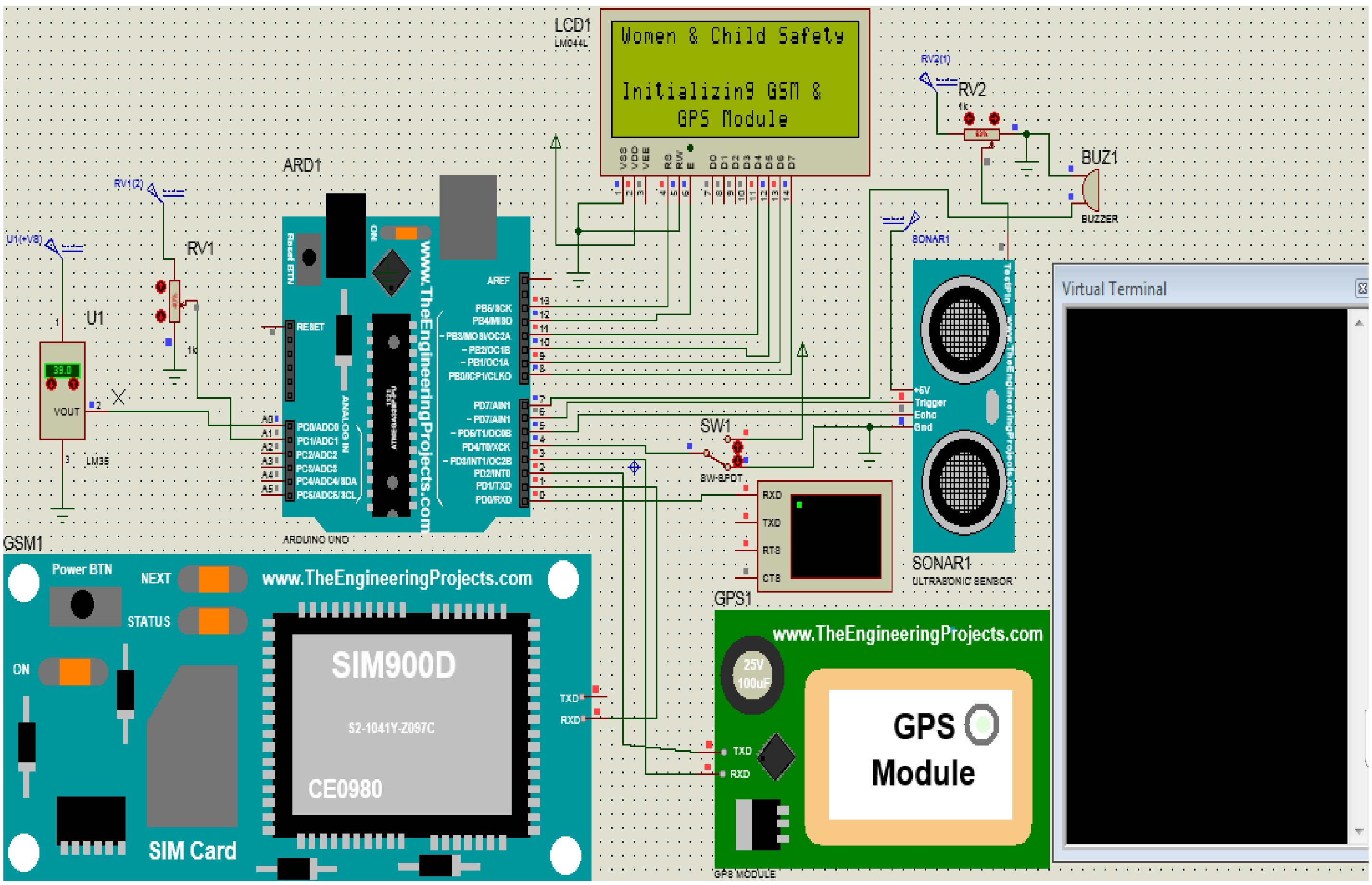

Fig 4.1 When project is turned ON

The above fig. displays the initialization of the project. When the project is turned on & the switch SW1 is in Women Safety Mode, then “Women & Child Safety – Initializing GSM & GPS Module” is displayed on the LCD and the GPS & GSM module is also initialized.

4.1.2. Women & Child Safety Mode :

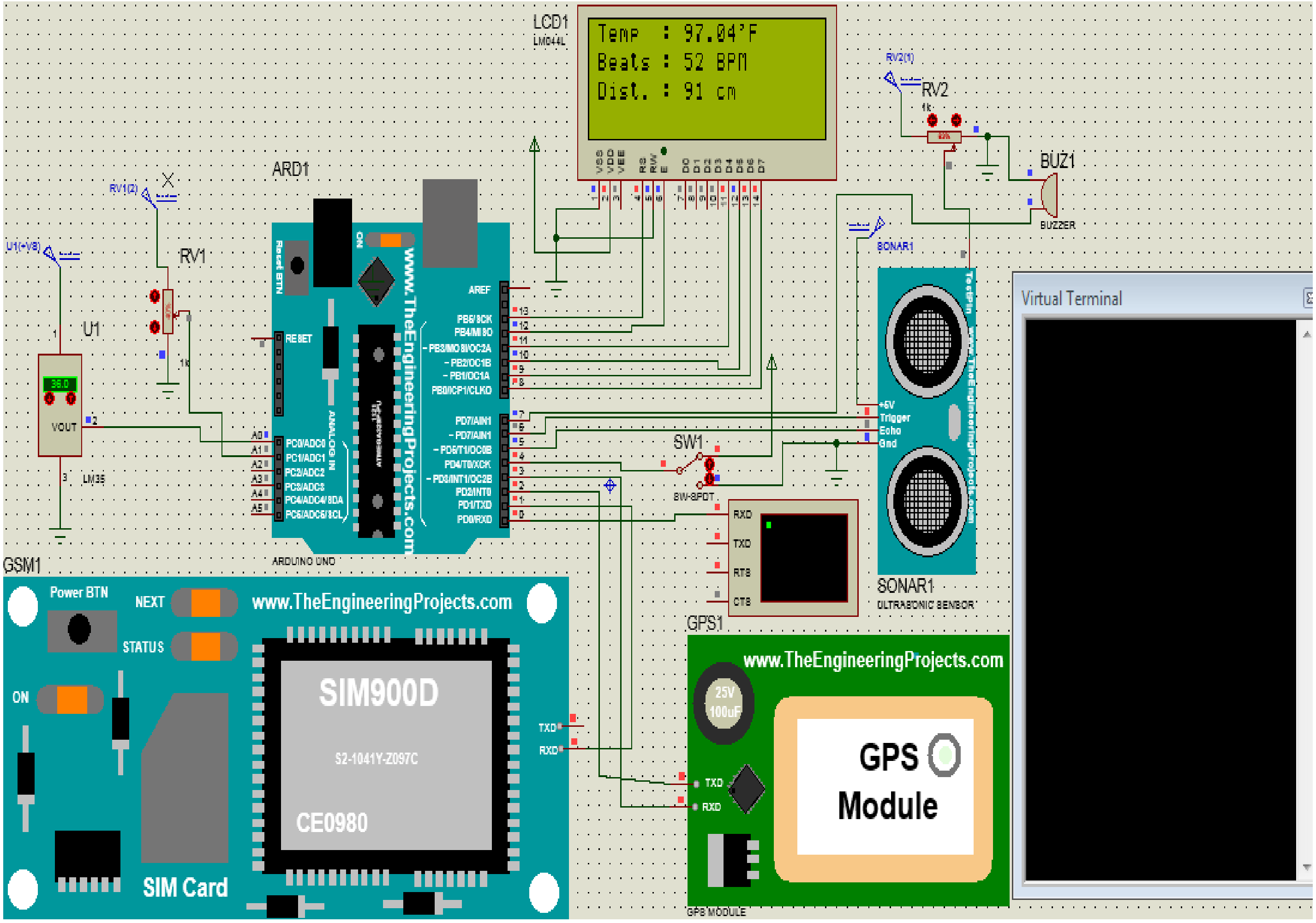

Fig 4.2 Simulation of Women & Child Safety(Normal Condition)

After the initialization of the project, all the sensors will start sensing the parameters such as Temperature, Heartbeats, Distance. In the above fig. temperature & distance is in the desired range but only heartbeat is below the desired level. So, as only one parameter is not in the threshold value then no action will be taken. Because as two sensors value will cross the threshold, then a message will be sent on the mobile number which is fed in the program.

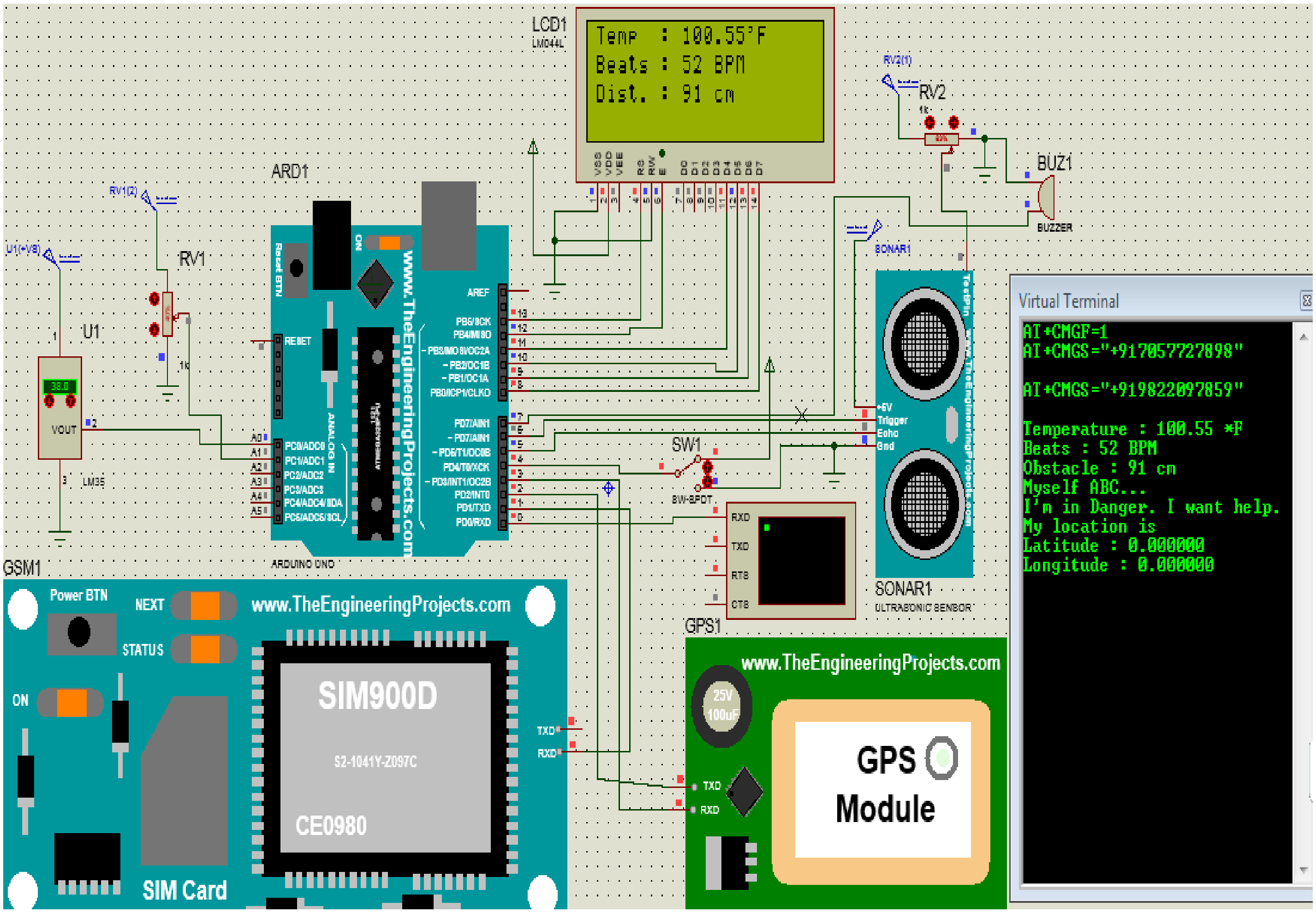

Fig 4.3 Temp. HIGH & Pulses LOW

In the above fig., two sensors values have crossed the threshold value i.e temperature value is above 100 & pulses (heartbeats) is below 60. So, a message will be sent to the registered mobile number. The message will be the parameter values & "Myself ABC…. I'm in Danger. I want help" & the location coordinates of the particular person who is being harassed.

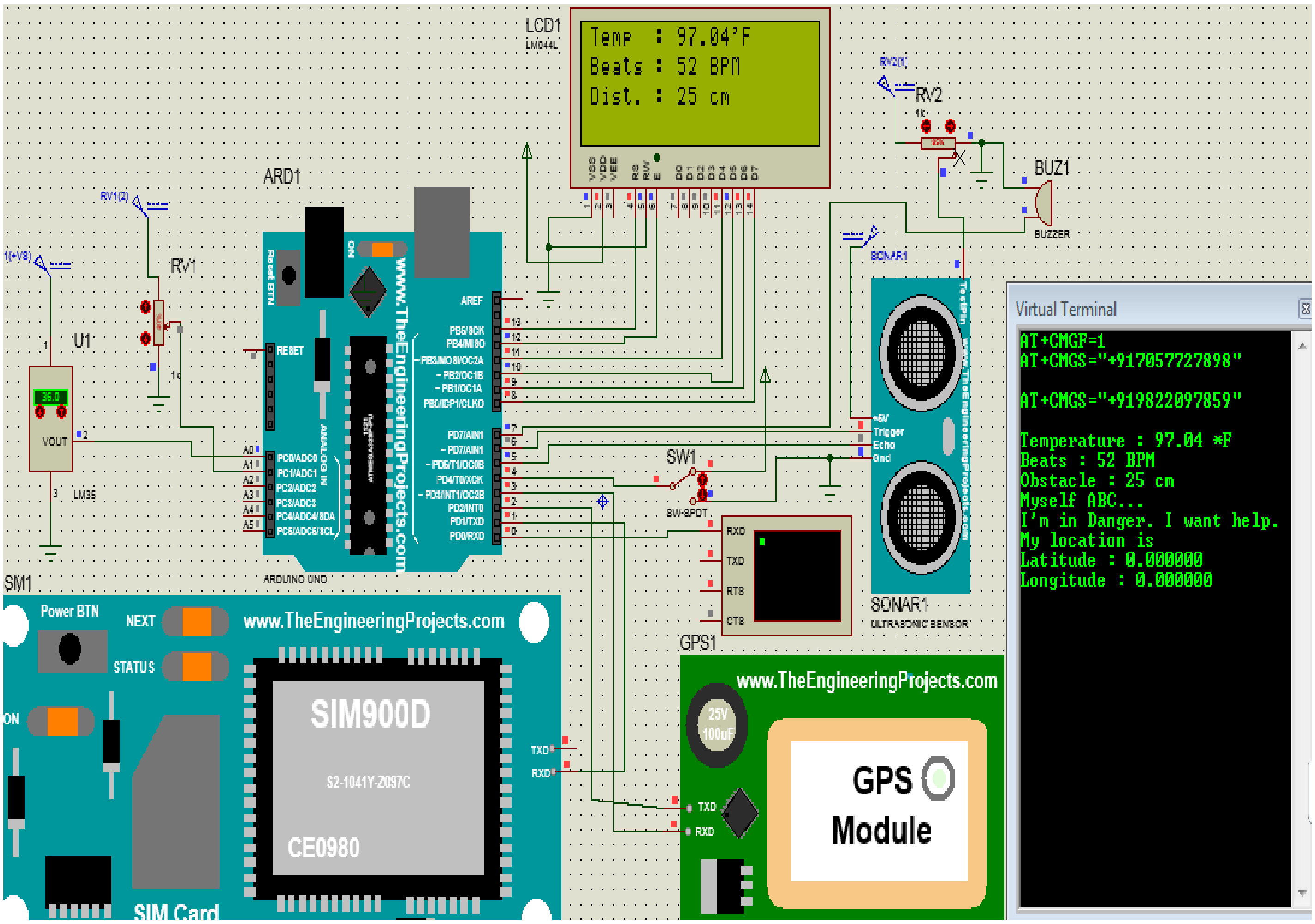

Fig 4.4 Pulses LOW & Short Distance

In the above fig., two sensors values have crossed the threshold value i.e Distance is below 50 & pulses (heartbeats) is below 60. So, a message will be sent to the registered mobile number. The message will be the parameter values & "Myself ABC…. I'm in Danger. I want help" & the location coordinates of the particular person who is being harassed.

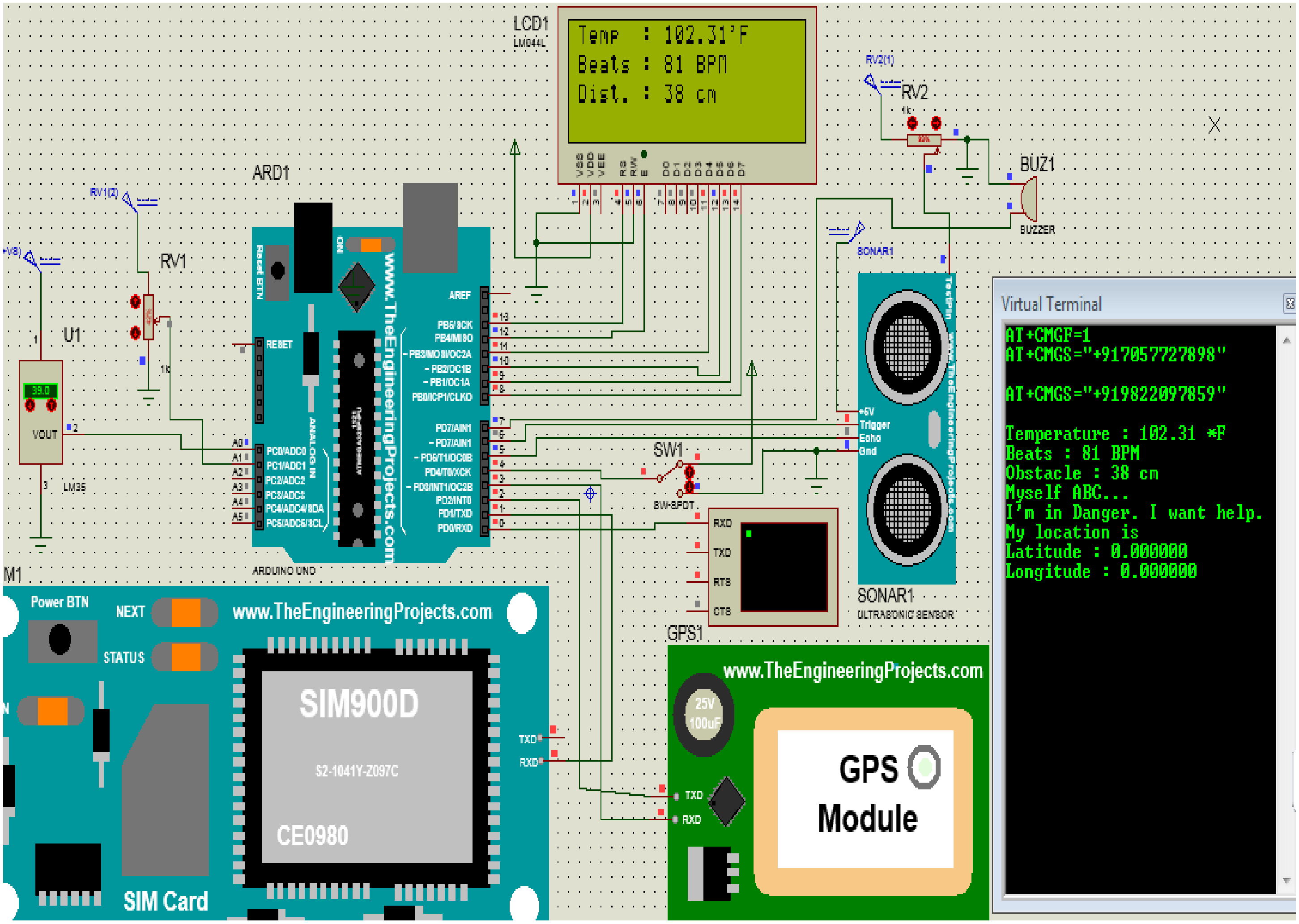

Fig 4.5 Temp. HIGH & Short Distance

In the above fig., two sensors values have crossed the threshold value i.e temperature value is above 100 & distance is below 50. So, a message will be sent to the registered mobile number. The message will be the parameter values & "Myself ABC…. I'm in Danger. I want help" & the location coordinates of the particular person who is being harassed.

4.1.3. Blind Person Safety Mode :

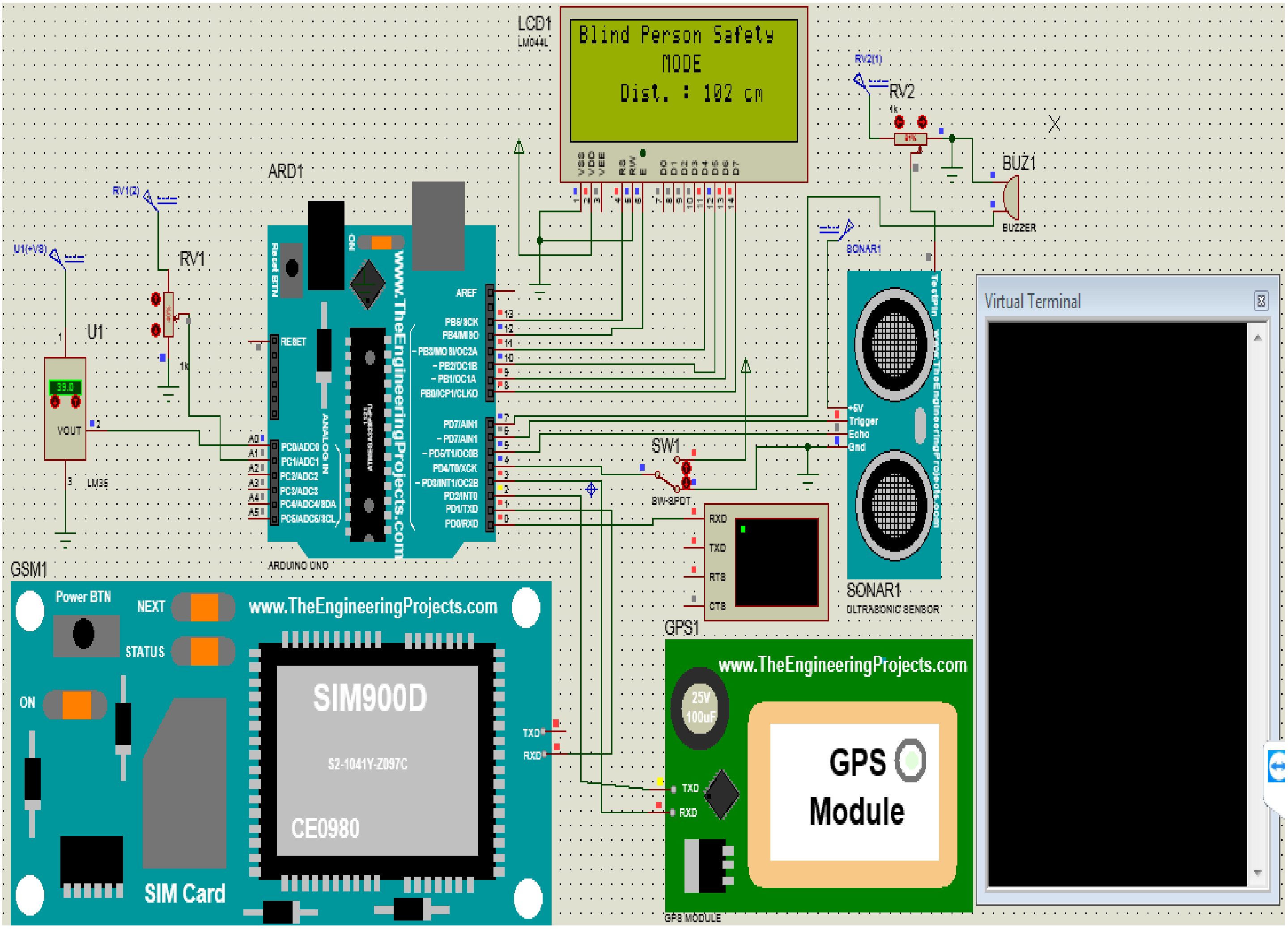

Fig 4.6 Simulation of Blind Person Safety (No Obstacle Detected)

The above fig. shows the project in Blind Person Safety Mode. As the switch SW1 is changed from ON to OFF then the mode of the project will be changed from Women Safety to Blind Person Safety. In this mode, if the distance is below 100cm then Buzzer will be turned ON, which will indicate that the obstacle is in front of the particular person. But in the above fig., distance is above 100, so it will take no action.

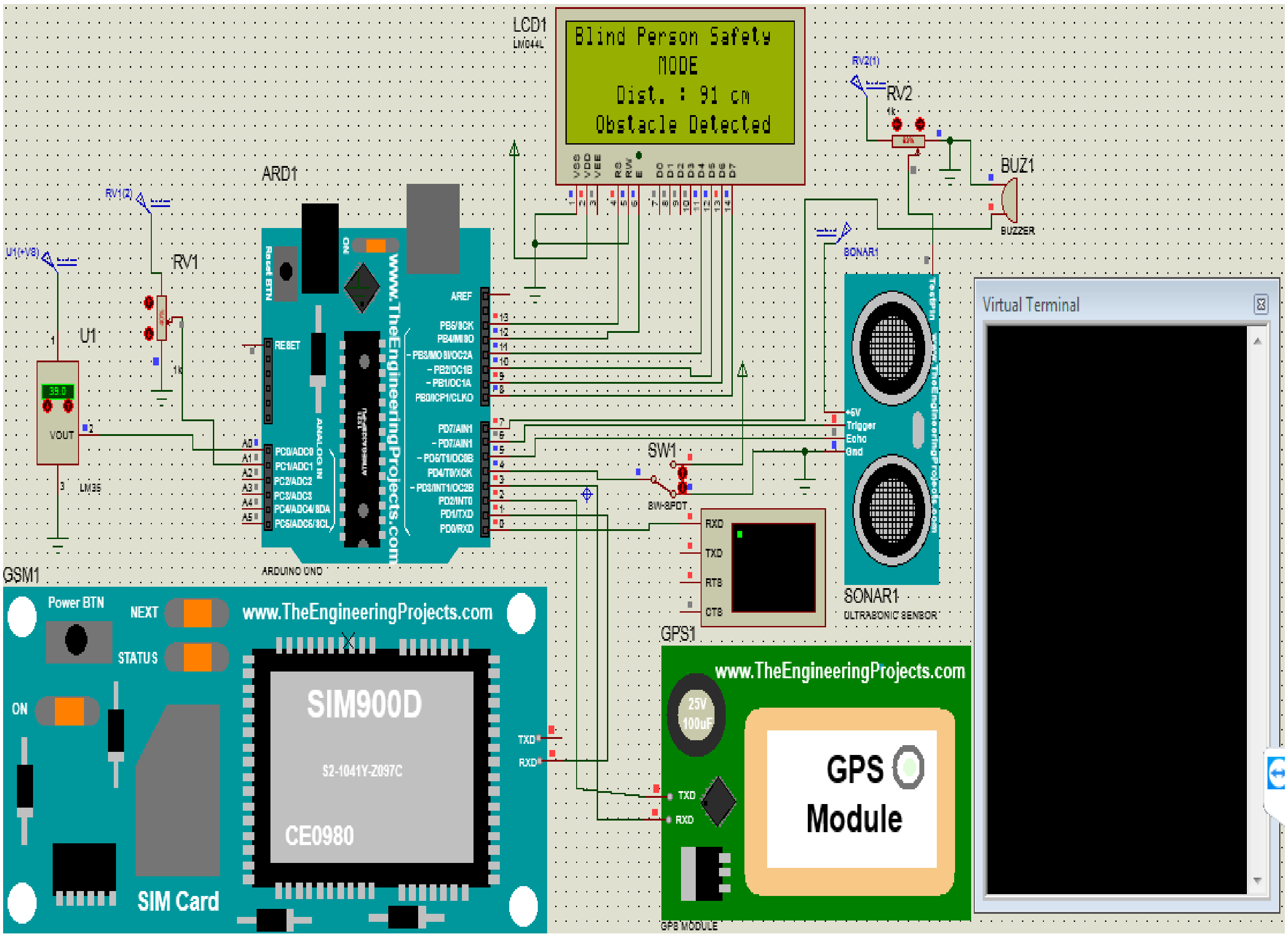

Fig 4.7 Simulation of Blind Person Safety (Obstacle Detected)

In the above fig., the distance is below 100, so as the sensor crosses the threshold value, the Buzzer will be turned ON & it will indicate that some obstruction is in front of the particular person & also message will be displayed on the LCD screen i.e. “Distance of the Obstacle & Obstacle Detected”.

4.2. Problems faced during Simulation :

1) We first decided to use Arduino Mega 2560 Controller for our project, but when we tried simulation with that controller we were not able to vary the values of the sensors, the we searched everywhere about that & we got to know that in Proteus Arduino Mega does not works properly. Then we decided to take Arduino UNO as our controller.
2) We used flex sensor in the project, but we are facing a problem in the simulation of flex sensor, because we can't see the exact value of Flex Sensor.
3) We also faced problem while sending the message through GSM.
4) We also faced problem with heartbeat sensor, it was not displaying any value, so we replaced it by the potentiometer, because heartbeat sensor does not gives exact value in simulation.
5) As there was issue with Flex Sensor and as the Flex Sensor & Flame Sensor has not much importance in the project, so we removed both the sensors.
6) As we were not able to see the actual values of sensors, so we interfaced a 20x4 LCD in the Simulation which will not be interfaced in the Hardware.

5. Conclusion and Future Scope

5.1. Conclusion based on Result

We have designed a Intelligent Security System for Women & Child Safety which also includes the Blind Person Safety using low-cost components, and included the message facility due to which the relatives / friend, etc. will be known about the location & the situation of the subject, due to which they can give the subject an immediate backup or help. As the system is small & will be packed in a small box, it will be easy to carry anywhere. Also it will help the blind person for avoiding him/her from getting harm. Also, this system will deal with critical issues faced by the women & will help to solve them with technologically message facility & ideas.

The following are some future goals for improvement :

5.1.1. Design Improvement :

We can increase the efficiency of the project and make it more powerful by adding some of the sensors such as IR Sensor & Flex Sensor which will be suitable with our idea.

5.1.2. Size :

We can reduce the size of the system and make it less complex by using single Modem which contains both the GSM & GPS facility on a single board due to which it will be easy to carry anywhere but it will increase the cost of the project.

5.2. Future Scope

1) Women Safety : As the Women feels very unsafe while going outside & also her parents feel tensed when she goes anywhere & getting late, this module will make the Women feel safe.
2) Primary School Children Safety : As the school children safety are major concerns for parents as well as school management due to the recent incidents of child crimes like children missing, abuse, kidnapping, etc. This module monitors the child safety when they are traveling in school buses.

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